

KRISHNA Koushik Thokala

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Summary

AWS-certified Solutions Architect and MS in Computer Science who builds AWS and Kubernetes infrastructure with Terraform, GitOps, and CI/CD — across EKS delivery, Prometheus/Grafana observability, and HPA autoscaling, validated on short-lived clusters. Backed by 2 years of 24/7 carrier-grade incident response — leading 5-engineer shifts through triage, RCA, and SLA-driven restoration for 25+ Tier-1 clients.

Certifications

AWS Certified Solutions Architect – Associate | Amazon Web Services | 2026

Technical Skills

Cloud Infrastructure: AWS, VPC, IAM, EKS, ECR, RDS, Lambda, DynamoDB, API Gateway, ALB, Terraform (IaC)
Kubernetes & Delivery: Kubernetes, Docker, Helm, Argo CD, GitOps, GitHub Actions, CI/CD, RBAC
Reliability & Observability: CloudWatch, Prometheus, Grafana, HPA, k6, on-call incident response, RCA, SLOs, MTTR
Programming & Systems: Python, Bash, SQL, PostgreSQL, FastAPI, REST APIs, Linux
Networking: TCP/IP, BGP, MPLS, DNS, Security Groups, NetworkPolicies

Projects

CloudOps GitOps Platform | EKS, Argo CD, Helm, Terraform, GitHub Actions **GitHub**

- Engineered an EKS GitOps platform where 4 Argo CD Applications reconcile Git-defined state across dev, staging, production-style, and observability namespaces; applied RBAC and NetworkPolicies and verified drift self-healing and Git-revert rollback.
- Provisioned and tore down the VPC, EKS, ECR, and IAM foundation with Terraform (20 resources); implemented immutable commit-SHA promotion and a branch-scoped GitHub OIDC role for ECR.

CloudOps SRE Platform | EKS, Terraform, Helm, FastAPI, React, RDS **GitHub**

- Built an EKS operations dashboard centralizing service health, deployments, incident timelines, and MTTR on a FastAPI, React, and PostgreSQL stack backed by RDS.
- Packaged the service with Helm-managed probes, resource limits, and ALB Ingress; secured RDS access with Secrets Manager credentials and restricted security groups and added structured JSON logging.
- Validated HPA behavior under k6 load as the backend scaled from 2 to 6 pods and returned to 2, processing 2,035 requests at 99.5% success.

CloudOps Uptime Monitor | Lambda, DynamoDB, API Gateway, Terraform **GitHub**

- Designed a serverless uptime monitor where EventBridge invokes scheduled Lambda checks, DynamoDB stores status history, SNS sends state-change alerts, and a CloudFront-hosted React dashboard displays live state.
- Modeled a latest-status access pattern that reduces records evaluated from 86,400 historical entries to 10 current-status records per lookup, avoiding full-history scans as retention grows.
- Ran a controlled outage drill validating 1-second failure detection, dashboard DOWN visibility within 26 seconds, duplicate-alert suppression, and recovery notification.

Experience

Tata Communications **July 2022 – July 2024**

Senior Engineer, Network Operations Center – Shift Lead *Pune, India*

- Led 5-engineer shifts in a 24/7 on-call NOC, owning a queue of 40+ tickets per shift across 25–30 Tier-1 carrier clients.
- Drove high-severity incident response end to end – triage, fault isolation, escalation, and restoration – resolving complex failures to root cause through structured log, alarm, and segment-level analysis.
- Sustained 99.9% service availability and a 4-hour restoration SLA across the client base by prioritizing P1/P2 incidents, driving timely resolution, and maintaining proactive customer communication.
- Authored post-incident RCAs and structured shift handoffs to drive recurrence prevention and continuity; earned a Certificate of Excellence for resolving an escalated incident into a documented customer win.

Education

Indiana University Bloomington **August 2024 – May 2026**

Master of Science in Computer Science, GPA: 3.5/4.0 *Bloomington, IN*

SRM Institute of Science and Technology **June 2018 – June 2022**

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.43/10 *Chennai, India*